

MudWatt Bioelectric League

Can your team make mud generate electricity and defend your design with data?

Win condition: Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation.

THE SCIENCE YOU ARE USING

Electrogenic bacteria. Some bacteria already in ordinary soil are electrogenic: as they digest nutrients, they push spare electrons onto surfaces outside their cells. Bury a graphite anode in oxygen-free mud and these microbes coat it, feeding it a steady stream of electrons.

Completing the circuit. Electrons leave the anode, run through the wire, the 100 kohm resistor, and the multimeter, reaching the cathode in the air on top, where they join oxygen and protons to form water. More microbes mean a higher voltage on the multimeter. Power is roughly voltage times current.

HOW TO BUILD AND RUN IT

- 01 Inoculate the mud.** Gloves on. Pack the vessel with sponge-damp soil. The bacteria you need are already in the dirt, so no culture is required.
- 02 Place the electrodes.** Bury the anode near the bottom and smooth mud over it so no air reaches it. Rest the cathode flat on top, exposed to the air.
- 03 Add a measured fuel.** Mix in a small, measured amount of one approved fuel. Record exactly how much. This is the food that drives the voltage.
- 04 Connect and read a baseline.** Clip the dry felt tabs to a 100 kohm resistor, set the multimeter to DC millivolts, and read the voltage across the resistor. Day one is often weak; the biofilm needs time.
- 05 Control one variable.** Pick ONE thing to test all week: fuel amount, moisture, electrode contact area, or mud source. Change only that, predict the effect, and compare to baseline.

Safety: Gloves required. No food near mud station. Leave the lid off so the cathode stays in air. Wash hands after handling soil. Allergy: Do not use nut-containing food scraps. Preferred fuels are sugar water; optional collected leaf litter.

MATERIALS

Carbon felt electrodes (cut as a flag with a dry tab)	2 per cell
24 oz deli cups	1 per cell
100 kohm resistor	1 per cell
Alligator leads	2 per cell
Fresh soil or mud	fill each vessel
Nitrile gloves	per student
Fuel	sugar water
Labels and tape	shared
Multimeters	required, 1 per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

Baseline result: _____

After redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Peak voltage (mV)	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	100